STOCKSENSE

The State of *Your Supply Chain.*

Automated demand forecasting, supplier outreach, and procurement intelligence — orchestrated by n8n.

Retailers lose money to *stockouts they could've predicted.*

$1.7T

ANNUAL GLOBAL LOSS

Retailers lose roughly $1.7 trillion every year to stockouts and overstock — preventable inventory misalignment.

8%

AVERAGE OUT-OF-STOCK RATE

Even disciplined retailers run out of items 8% of the time. In SMEs, the figure regularly exceeds 15%.

Manual

STILL THE NORM

Most SMEs forecast demand on gut feel and contact suppliers reactively — long after a stockout has already hurt sales.

And the data to predict it has been sitting in their spreadsheets all along.

Three workflows.
One supply chain, on autopilot.

WF-1

Demand Forecasting

Reads historical stock data. GLM-5.1 predicts next-period demand per product per region — confidence scored, stockout-flagged.

WF-2

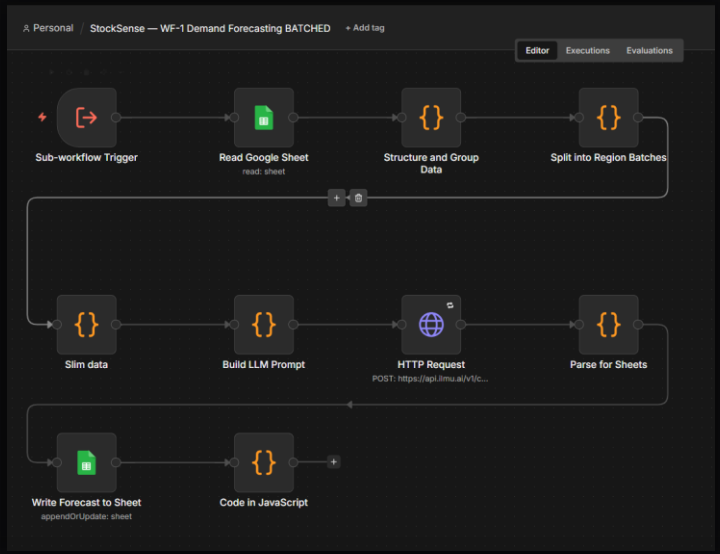
Supplier Outreach

When risk is flagged, drafts and sends personalised order-request emails to the right suppliers. Idempotent — never spams.

WF-3

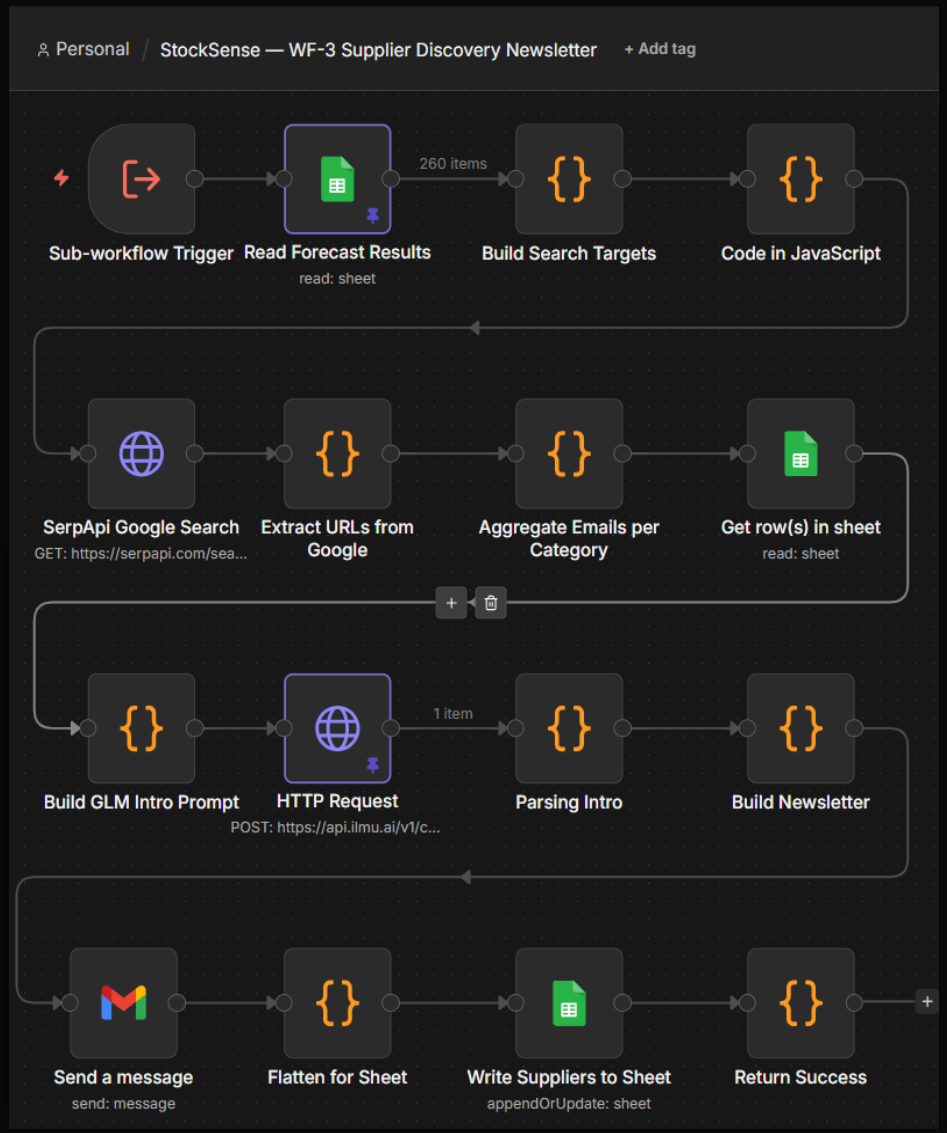
Supplier Discovery

Weekly: scans the web for new Malaysian suppliers, validates contact emails, and emails a curated digest to the supply manager.

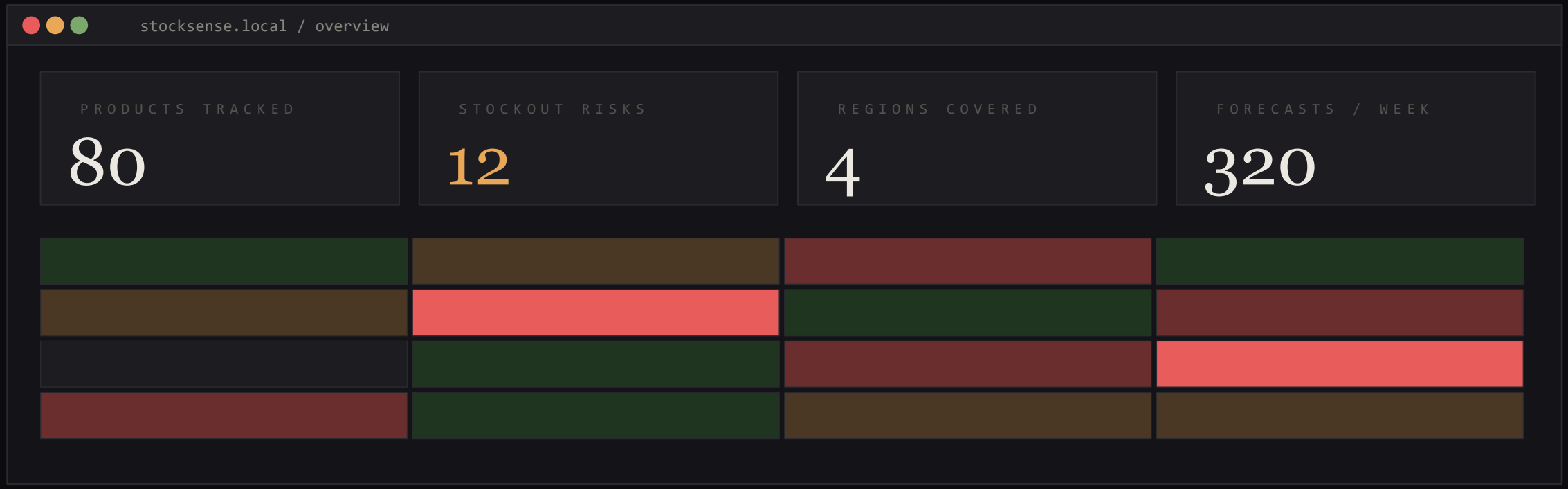

Personal / StockSense — WF-1 Demand Forecasting BATCHED
+ Add tag
Editor
Executions
Evaluations
Sub-workflow Trigger
Read Google Sheet
read: sheet
Structure and Group Data
Split into Region Batches
Slim data
Build LLM Prompt
HTTP Request
POST: https://api.ilmu.ai/v1/c...
Parse for Sheets
Write Forecast to Sheet
appendOrUpdate: sheet
Code in JavaScript

STOCKSENSE

Personal / StockSense — WF-2 Supplier Contact
+ Add tag
0 / 3
Published
Editor
Executions
Evaluations
Sub-workflow Trigger
Read Contact Log
read: sheet
Read Forecast Results1
read: sheet
260 items
Read Supplier Contacts
read: sheet
64 items
Match Urgent to Suppliers
Code in JavaScript1
Should Skip?1
true
false
Return Skipped1
Return Success1
Code in JavaScript
Parse Email Draft1
Send Email to Supplier
send: message
Data Compile
Log to Sheet1
appendOrUpdate: sheet


Personal / StockSense — WF-3 Supplier Discovery Newsletter
+ Add tag
Sub-workflow Trigger
Read Forecast Results
read: sheet
260 items
Build Search Targets
Code in JavaScript
SerpApi Google Search
GET: https://serpapi.com/sea...
Extract URLs from Google
Aggregate Emails per Category
Get row(s) in sheet
read: sheet
Build GLM Intro Prompt
HTTP Request
POST: https://api.ilmu.ai/v1/c...
1 item
Parsing Intro
Build Newsletter
Send a message
send: message
Flatten for Sheet
Write Suppliers to Sheet
appendOrUpdate: sheet
Return Success

Everything in *one editorial-grade dashboard.*



Overview · Triggers · Forecasts · Contacts · Discovered — five pages, real-time, deployable.

How it all *hangs together.*

FRONTEND

Next.js 15

- Overview
- Triggers
- Forecasts
- Contacts
- Discovered

BACKEND

n8n workflows

- 6 webhook APIs
- WF-1 Forecasting
- WF-2 Outreach
- WF-3 Discovery
- Master orchestrator

DATA & SERVICES

External integrations

- Google Sheets (DB)
- ILMU GLM-5.1
- Gmail API
- SerpApi (search)

Frontend talks to n8n via 6 webhooks. n8n executes workflows. Workflows hit external services. Data lives in Google Sheets.

Why n8n as a *backend?*

Most hackathon teams burn 60% of their build time on backend plumbing. We chose n8n so we could ship observable AI pipelines instead.

Visual debugging — Every LLM call, sheet write, and email send is inspectable in the execution log.

Zero boilerplate — Google Sheets + Gmail + HTTP integrations wire up in minutes, not days.

Observable AI pipelines — When GLM-5.1 hit Cloudflare timeouts, we could see exactly where and why.

Portable workflows — The whole system ships as 10 JSON files anyone can import. No deploy pipelines.

What we built *in 48 hours.*

10 n8n WORKFLOWS	**1,600** LINES OF FRONTEND	**5** DASHBOARD PAGES
80+ FORECASTS / RUN	**4** REGIONS COVERED	**0** BACKEND SERVERS

End-to-end automation, fully working, with editorial-grade UI and zero custom backend code.

Things that broke, *and how we fixed them.*

PROBLEM	SOLUTION
ILMU's 60s Cloudflare hard timeout	Sequential batching — 5 products per call, retry-on-fail with 10s backoff.
GLM-5.1 sometimes returns malformed JSON	Strict JSON-mode prompting + defensive parser with fallback responses.
Email scraping from random supplier websites	Discovered SerpApi's snippet_highlighted_words already contains validated emails — skipped HTML scraping entirely.
CORS between Next.js and self-hosted n8n	Routed every call through Next.js API routes as a server-side proxy.

Where we *take it next.*

NOW

Hackathon MVP

10 workflows, dashboard, demo data. Manual triggers.

+1 MONTH

Production hardening

Postgres replaces Sheets. Auth via NextAuth. Real-time SSE workflow status.

+3 MONTHS

Multi-tenant

Multiple businesses, each with their own data scope and supplier book.

+6 MONTHS

Marketplace

Mobile companion app. Auto-negotiation flows. Supplier rating system.

Thank you.

Now go restock the shelves.

PROJECT	DEMO	STACK
github.com/schizoalloy/stocksense	*Stocksense demo video*	*Next.js 15 / n8n / GLM-5.1*